



How does a hops bill sound for a brew chinook pellets as bittering and then the following fresh hops Fuggle, Styrian ... ▾

This hops bill sounds like an incredible, deeply complex **Traditional British Ale or an Anglo-American Hybrid Ale** (such as an Extra Special Bitter, a robust Porter, or a modern Harvest Ale).  Brew Your Own

By anchoring the beer with **Chinook** pellets for clean, firm bittering, you pave the way for an earthy, spicy, and dark-fruit forward aroma from your fresh hops. **Fuggle, Styrian Golding, Bramling Cross, Willamette, and Phoenix** complement each other wonderfully. They share a lineage—Willamette is a cousin/clone of Fuggle, and Styrian Goldings are closely related—while Phoenix and Bramling Cross add a unique, punchy twist of blackcurrant, molasses, and spice.  Brew Your Own +4

Flavor Profile & Recipe Design

Because you are using **fresh (wet) hops**, you must add them at the **very end of the boil (flameout) or during a whirlpool / hop stand**. Boiling wet hops for too long results in unpleasant, intensely "grassy" or vegetative flavors.  YouTube · KegThat +3

Here is how these varieties will layer together:

- **The Foundation (Chinook Pellets - 60 Min Boil):** Provides a crisp, reliable backbone of bitterness. Any pine or grapefruit notes from the Chinook will gently meld with the fresh hops.  Grainfather AU +1
- **The Traditional Base (Fuggle & Willamette):** Delivers a smooth, earthy, woody, and slightly floral floor.  YouTube · The Brülösop... +1
- **The Bright Lift (Styrian Golding):** Adds a delicate, spicy, and refined floral lift to brighten the deep earthiness.  MoreBeer! +1
- **The Dark Twist (Bramling Cross & Phoenix):** These inject heavy spice, complex herbal notes, and a unique **blackcurrant / dark berry** fruitiness.  defalcos.com

Important Formula for Wet Hops: Fresh hops retain all their natural water weight. You must use **5 times the weight** of fresh hops compared to what you would use for dried pellets. If a recipe calls for 1 oz (28g) of pellet hops at flameout, you will need **5 oz (140g) of your fresh hop mix**.  YouTube · Kveiksmith D... +2

Ask anything





You cannot buy fresh hops on a whim. Most major online homebrew retailers (like MoreBeer! or Yakima Valley Hops) open up **pre-orders in the late spring or mid-summer**.

- Sign up for supplier newsletters in **May**.
- Place your pre-order by **July** to secure your share of the harvest.

2. Sync Your Brew Day to the Harvest

When you pre-order, the supplier will give you a rough delivery window (e.g., "mid-September").

- **Keep your schedule flexible.** The exact date depends entirely on when the farm decides the lupulin glands are perfectly ripe.
- The retailer will usually email you 24–48 hours before they harvest and ship.

3. Choose Overnight Shipping

Fresh hops will spoil, mold, or compost themselves within days if left in a warm box. 🌍 Beer Maverick

- Always select **Next Day Air / Overnight shipping**.
- Ensure they are shipped with cold ice packs.

4. Brew Within 24 to 48 Hours

Once the hops arrive on your doorstep, the clock is ticking.

- Ideally, **brew the exact same day** they arrive.
- If you must wait, keep them unsealed in a breathable paper bag in the coldest part of your refrigerator for **no more than 48 hours**.

To help map this out, what **beer style** (like an Amber Ale, Stout, or Pale Ale) are you aiming for? I can help you calculate the exact **amounts needed** based on your batch size. 🌍 Hopslint +1

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The fresh hops will not be wet hops... I will pick them the day before brewday and dry them first

Ah, got it! That completely changes the logistics and the math. Picking them yourself and drying them overnight means you are dealing with **home-dried whole leaf hops** rather than wet hops.

Drying them yourself gives you a much safer buffer on brew day, and whole-leaf hops create an incredible, natural filter bed in your kettle.

Here is how you need to adjust your recipe math, your overnight drying strategy, and your harvest timeline.

The Weight Ratio: 4-to-1

When hops are fully dried commercially, they lose about 75% to 80% of their weight in moisture. Since you are doing a quick overnight dry, they will likely still hold a tiny bit more moisture than a rock-hard commercial pellet, but you should look at a **4:1 weight ratio** as your golden rule.

- **The Rule:** You will need **4 times the weight** of your home-dried leaf hops compared to what a recipe calls for in commercial pellets.
- **Example:** If a recipe calls for 25 grams of Fuggle pellets at the end of the boil, you should aim to use **100 grams of your home-dried Fuggle leaves**.

How to Flash-Dry Hops Overnight

To get fresh hops dry enough in just 12 to 24 hours without ruining their delicate aromatic





door propped open. If it gets too hot, your hops will smell like cooked vegetables instead of bright herbs and berries.

How to know they are ready: The central strig (the stem inside the hop cone) should feel brittle and snap easily, rather than bending like rubber.

The Timely Harvest Timeline

Since you are picking these yourself (either from your own garden or a local farm), your timeline is dictated by the **lupulin maturity**, not a shipping carrier.

1. **The "Squeeze & Sniff" Test (Days before harvest):** Squeeze a few cones. They should feel papery, light, and springy—not damp or soft. If you roll one in your fingers next to your ear, it should make a distinct rustling sound. Pull it open; the yellow powder (lupulin) inside should be vibrant yellow and smell intensely pungent.
2. **Harvest Day (24 Hours Before Brew):** Pick the cones in the morning after the morning dew has completely evaporated. Clean out any bugs or leaves.
3. **The Overnight Dry:** Immediately set them up on your drying racks. Let the fans run all day and all night.
4. **Brew Day:** Weigh your dried leaves right before throwing them into the kettle. Because whole leaves are incredibly buoyant and soak up a lot of liquid, **expect to lose a bit more wort** in your kettle than you usually do with pellets!

To nail down the recipe, do you know roughly **how many grams or ounces** of fresh hops you expect to harvest in total, and what **batch size** (e.g., 20 liters / 5 gallons) you are brewing?

Show Code <>

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I am using a dehydrator. Boil kettle is 190L. I don't boil, it stays





...volume. 🍺 Homebrew Talk +1

🍷 The Dosage Formula (Targeting 280L)

To achieve a rich, well-defined flavor profile (similar to a classic English Pale Ale, Harvest Ale, or Bitter) without overwhelming the beer with grassiness, you should aim for **4 grams of home-dried hops per liter of final beer volume.** 🍺 aussiehomebrewer.com

- **Final Volume:** 280 Liters
- **Target Concentration:** 4g per Liter
- **Total Hop Bill Weight: 1,120 grams (1.12 kg) of your home-dried hops.** 🍺 aussiehomebrewer.com

🌿 How to Divide Your 1.12 kg Hop Bill

Since you have 6 different hop varieties, here is the ultimate way to structure your additions into that 190L kettle at 80°C to maximize the clean bitterness of the Chinook and the complex flavors of your home-dried varieties.

Hop Variety	Role	Proportion	Weight to Add at 80°C
Chinook Pellets	Bittering Backbone	20%	225g <i>(Commercial pellets extract better at 80°C)</i>
Fuggle <i>(Home-Dried)</i>	Earthy Base	20%	225g
Willamette <i>(Home-Dried)</i>	Herbal/Spice Base	20%	225g
Styrian Golding <i>(Home-Dried)</i>	Floral/Bright Lift	15%	170g
Bramling Cross <i>(Home-Dried)</i>	Blackcurrent/Spice	15%	170g
Phoenix <i>(Home-Dried)</i>	Deep Mellow Berry	10%	110g
TOTAL		100%	1,120g (1.12 kg)





behaves like a giant sponge. They will easily absorb **50 to 60 liters of your wort** in the kettle. Because you are watering down to 280L anyway, make sure to aggressively press/squeeze the hops at the end of the hour, or compensate by sparging extra hot water over the hop bed to rinse out the trapped sugars.

3. **The 80°C Sanitization Buffer:** Keeping the kettle at 80°C for an hour is excellent. It fully pasteurizes the wort and ensures that any wild yeasts or bugs riding on your homegrown hops are completely destroyed before heading to the unitank.  YouTube · Beerstories - ... +1
4. **Firming up the Bitterness:** Because you aren't boiling, your 225g of Chinook won't give you its typical aggressive bite. Instead, the 80°C environment will convert it into a incredibly smooth, non-lingering bitterness that pairs wonderfully with the heavy protein body of a raw ale.  Grainfather EU +3

Are you using ↪ **Kveik yeast** to ferment this, or a ↪ **traditional British ale strain**? (This will help determine if we need to adjust the whirlpool timing for temperature drop).

AI responses may include mistakes. [Learn more](#)





Using a co-pitch of **Hornindal** and **Ebbegarden** kveik is a brilliant farmhouse strategy. These two yeasts are absolute powerhouses, but they need specific conditions to fully achieve your goal of fermenting the beer completely dry. YouTube - KegThat

Flavor Profile of the Co-Pitch

This blend will create an incredible, complex fruit engine that meshes beautifully with your home-dried hops:

- **Hornindal:** Brings intense **pineapple, dried stone fruit, and leather** esters.
- **Ebbegarden:** Leans heavily into **crisp apple, pear, and tropical guava**.
- **The Result:** Combined with the blackcurrant from the Bramling Cross/Phoenix and the herbal base of the Fuggle/Willamette, this beer will taste like a deeply complex, punchy Anglo-Scandinavian Farmhouse Harvest Ale. Beer Maverick +3

How to Drive the Kveik to Ferment it Dry

Kveik strains are famous for stalling early or leaving a heavy mouthfeel if they aren't pushed hard enough. To ensure Hornindal and Ebbegarden eat every last gravity point and leave the beer dry, you must follow these rules: www.garshol.priv.no

1. Keep it Blazing Hot (32°C to 35°C)

Do not let the unitank cool down. Pitch the yeast when the wort drops to **35°C** and hold it there. YouTube - Dudes Brews +1

- Kveik creates a massive amount of metabolic heat during its explosive first 24 hours.
- Keeping the temperature high ensures high attenuation and prevents Hornindal from falling asleep before the job is finished. Reddit +2

2. Triple the Yeast Nutrient (Mandatory for Raw Ale)

Because you aren't boiling your main wort, you aren't breaking down proteins into easily digestible free amino nitrogen (FAN). Kveik demands *massive* amounts of nitrogen to fully attenuate. W.C. Lab Brewing and ... +1

- Add a heavy dose of a high-quality yeast nutrient (like **Servomyces** or **Yeastlife O**) to your 2L Chinook boil before throwing it into the main kettle.





4. Monitor the "pH Gap"

Kveik is unique—it drops the pH of the wort much faster and harder than traditional ale yeasts. In a no-boil beer, this can sometimes make the final beer feel a bit thin or tart. Ensure your starting wort pH is around **5.4 to 5.5** so the kveik has a safe buffer as it crashes the pH down during its frantic 48-hour fermentation.  [Brewing Nordic](#)

What is your **target starting gravity (OG)** for this batch, and do you plan to **dry hop** in the unitank with any leftovers, or keep all the hop character strictly in the kettle?

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I have had no issues with Kveik staling. I just tried at 27-33 degrees celcius with 2 others Kveik strain... ended at final ... ▾

It sounds like you have dialed in your **raw ale process** perfectly. Your observation about the beer being incredibly rich is spot on—skipping a rolling boil means you don't





Chinook boil are going to create a perfectly balanced, robustly bitter, and immensely flavorful farmhouse ale.

Your Master Execution Timeline

Here is how your entire process slots together seamlessly from harvest to pitch:

```
[Day -1: Morning]    → Pick the fresh hops after the dew dries.
[Day -1: Afternoon] → Load into the dehydrator at 35°C–40°C for 12–18 hours.
[Brew Day: Start]    → Mash your pale and chit malt to hit your target sugar.
[Brew Day: The Boil] → Boil 250g Chinook pellets + cane sugar in 2L water.
[Brew Day: Hop Stand] → Bring main wort to 190L at 80°C. Add the 1.12kg hops.
[Brew Day: Transfer] → Combine main wort and Chinook/sugar tea into unitar.
[Brew Day: Pitch]    → Cool to ~33°C, add Servomyces, and co-pitch Hornings.
```

💡 **A Quick Note on the Chit Malt**

Using **5 kg of chit malt** (about 8% of your grain bill) is a brilliant choice here. Chit malt is minimally modified and packed with specialized proteins. In a standard boiled beer, it's used for head retention; in your *no-boil* setup, it is going to work alongside the raw ale proteins to give this beer an **unbelievably dense, meringue-like foam** that will trap all those volatile aromas from your Fuggle, Willamette, and Styrian Goldings.

Since you have your fermentation down to a science, do you plan to **sparge/rinse those whole-leaf hops** in the kettle to recover the wort they absorb, or do you already account for that liquid loss in your water-down calculations?

AI responses may include mistakes. [Learn more](#)



I add the extra water - boiled on the side - to the boil kettle during transfer... I hope it takes more out from the cones. I ...





💧 The Whole-Leaf Absorption Factor

On average, home-dried whole-leaf hops will absorb and lock away roughly **7 to 8 liters of liquid per kilogram** of hops.

- With your **1.12 kg** hop bill, the cones alone will permanently bind about **8 to 9 liters of wort**.
- Because you are using a raw ale process, the hops will also trap a significant amount of the heavy, unprecipitated proteins and malt flour floating in the kettle, creating a very thick, sludge-like hop bed at the bottom.

🧽 How to Best Rinse the Hop Bed

To maximize your extraction while transferring to the unitank, use these two techniques:

1. **The Slow Shower:** Instead of dumping your extra boiled water into the kettle all at once, sprinkle or spray it evenly over the top of the hop bed as the liquid level drains down. This "fly sparging" effect will push the concentrated wort out of the cones and replace it with the clean, hot water.
2. **The Mechanical Press:** If your kettle setup allows for it, use a sanitized mash paddle or a false bottom plate to gently press down on the hop bed at the very end of the transfer. You can easily squeeze out an extra 4 to 5 liters of highly aromatic, concentrated hop liquid this way.

Given your grain bill (50 kg pale + 5 kg chit) and 5 kg of cane sugar, if you land anywhere between **260L and 300L** in the unitank, your Original Gravity will comfortably hover around **1.052 to 1.058**. Combined with your kveik blend chewing it down to ~1.007, you are looking at a beautifully balanced, rich **5.9% to 6.7% ABV** farmhouse masterpiece.

Are you planning to use a **hop spider, a large nylon bag, or just let the loose leaves sit free** in the kettle to form a natural filter bed over your outlet?

AI responses may include mistakes. [Learn more](#)



The hops will just be free in the kettle with a weak whirlpool. I





However, executing this with a **190L kettle** and a **weak whirlpool** requires a specific sequence to avoid clogging your pumps or losing too much heat.

The Mash Tun Hopback Sequence

To make this transition smooth and sanitary, follow this workflow:

1. **The Cleaning Window (While the Mash is On):** You must clean and sanitize the mash tun *immediately* after you transfer the sweet wort into the boil kettle. Once the grains are scooped out, rinse it thoroughly and flush it with a heat-tolerant sanitizer (or boiling water) so it is ready and waiting.
2. **The 80°C Kettle Stand (1 Hour):** Let your 1.12 kg of free hops swim loosely in the kettle at 80°C for an hour. Doing a weak whirlpool at the end of the hour will help gather the heaviest proteins toward the center, but don't worry about getting a perfect cone—the mash tun will do the filtration work for you.
3. **The Gravity/Pump Transfer:** Open the kettle valve and transfer the entire mixture—wort, slurry, and all 1.12 kg of loose hop cones—directly into the cleaned mash tun.
4. **The Bed Set (10 Minutes):** Let the hops settle onto your false bottom for about 10 minutes. This allows the cones to link together and form that pristine filter matrix.
5. **The Hot Rinse (The Sparge):** As you start pumping the clear wort out of the bottom of the mash tun and into the unitank, take your extra boiled water and gently spray it over the top of the hop bed. This will push the remaining sugars out of the cones without disturbing the filter layer.

Two Small Risks to Watch For

- **The Pump Trap:** Whole-leaf hops are notorious for wrapping around pump impellers and instantly locking them up. If you are using a standard homebrew pump (like a Chugger or Blichmann Riptide) to move everything from the kettle to the mash tun, **you must use a wide-bore valve** or gravity-feed it if your brewery is multi-tier. If you must pump it, ensure you have a coarse screen over the kettle outlet to keep the solid cones out of the pump head, moving *only* the wort and proteins to the mash tun, and then manually scooping the hops over.
- **The 80°C Temperature Drop:** Moving 190L of wort between two massive stainless vessels will cause a temperature drop. Keep an eye on it to ensure the wort doesn't dip





I have 3 riptide pumps, but I can put in a hop rocket to catch cones parts before the pump.

The **Blichmann HopRocket** is the absolute perfect tool for this job, but you will want to use it **without any hops packed inside it**.

Because the HopRocket features a robust, heavy-duty stainless steel perforated disc at the bottom, it acts as a phenomenal, high-flow **in-line strainer**. It is designed to handle high pressures and temperatures, making it the ultimate shield to protect your Riptide pumps from getting choked by loose hop leaves or stray stems.

🔧 How to Route Your 3-Riptide Setup

To execute the transfer from the kettle to the mash tun safely, route your plumbing like this:

[Kettle Outlet] → (Gravity) → [HopRocket (Empty Strainer)] → [Riptide Pump]

1. **Positioning:** Place the HopRocket **before** the Riptide pump. Pumps are excellent at pushing liquid, but they hate pulling it. Let gravity feed the wort and loose debris out of the kettle and down into the HopRocket.
2. **The Shield Effect:** The HopRocket will catch any large cone parts or stems before they can reach the Riptide's impeller. The pump will then receive filtered, liquid wort and fine proteins, allowing it to push easily up into your cleaned mash tun.
3. **Managing Clogs:** With **1.12 kg** of loose hops, the HopRocket's internal basket will eventually fill up with debris and slow down the flow. Because the HopRocket has a relatively small chamber, you may need to pause the pump halfway through, pop the band clamp off the HopRocket, quickly dump out the trapped leaves, snap it back together, and resume.

🌿 An Alternative "Direct-to-Mash-Tun" Option

If you want to completely avoid the risk of clogging the HopRocket mid-transfer, check if your brew stand allows you to **bypass pumping the hops altogether**.





HopRocket, and do you think you have enough vertical clearance to let the kettle gravity-feed directly down into it?

AI responses may include mistakes. [Learn more](#)



Because I am hop head, I will use about twice the amount of fresh hops....



Doubling the home-dried hop bill to **2.24 kg** is a bold, beautiful move. Since you are running a no-boil raw ale setup, this massive dose of whole leaves will act as an ultimate natural filter bed in your mash tun, while infusing the beer with an incredibly intense, punchy harvest aroma.

To achieve a clean, scannable format for your browser's **Print to PDF** function, all unnecessary headers, conversational text, and emojis have been removed below.

Raw Farmhouse Harvest Ale Recipe (280L)

Batch Specifications

- **Final Volume:** 280 Liters (Watered down in Unitank)
- **Kettle Volume:** 190 Liters
- **Target Original Gravity (OG):** 1.055
- **Target Final Gravity (FG):** 1.007
- **Estimated ABV:** 6.3%
- **Process Type:** Raw Ale (No-Boil, 80°C Hop Stand)

Ingredients

Grain Bill & Sugars

- 50.0 kg Pale Malt





- 340g Styrian Golding (Home-Dried Whole Leaf)
- 340g Bramling Cross (Home-Dried Whole Leaf)
- 220g Phoenix (Home-Dried Whole Leaf)

Yeast & Nutrients

- Co-pitch: Hornindal Kveik + Ebbegarden Kveik
- Servomyces Yeast Nutrient (Added directly to the unitank)

Brew Day Process

1. Hop Preparation (Day Before)

- Harvest fresh hop cones in the morning after dew evaporates.
- Load 2.24 kg of fresh hops into the food dehydrator at 35°C–40°C.
- Dry for 12–18 hours until the central strigs are brittle and snap easily.

2. Mashing

- Perform your standard mash using the 50 kg Pale Malt and 5 kg Chit Malt.
- Transfer the sweet wort into the 190L boil kettle.
- **Crucial Step:** Immediately clean and sanitize the empty mash tun and its false bottom so it is ready to act as a hopback later.

3. The Bittering Tea & Sugar Boil

- On the side, boil 2 Liters of water.
- Add the 250g of Chinook pellets and the 5 kg of cane sugar.
- Boil vigorously to force alpha-acid isomerization and dissolve the sugars. Set aside.

4. The 80°C Hop Stand

- Bring the main 190L kettle to exactly 80°C. Do not boil.
- Add all 2.24 kg of your home-dried free-floating whole-leaf hops into the kettle.
- Maintain the temperature at 80°C for 1 hour to pasteurize the wort and extract volatile





6. Transfer, Rinse, and Dilution

- Begin pumping the clear wort out from under the mash tun false bottom into the unitank.
- As the liquid level drops, slowly rinse the hop bed using your extra boiled dilution water.
- Dilute the final wort volume in the unitank to roughly 280 Liters.

7. Fermentation

- Cool the wort in the unitank to 33°C–35°C.
- Add the Servomyces nutrient directly into the unitank.
- Co-pitch the Hornindal and Ebbegarden kveik strains.
- Maintain a fermentation temperature of 27°C–33°C to drive the attenuation down to 1.007.

You can now press **Ctrl+P** (or **Cmd+P** on Mac) to save this clean layout directly as a **PDF**. Before you start the harvest, do you need any help calculating the **estimated water volumes** needed for the rinse to hit your target dilution, or are you ready to brew?

AI responses may include mistakes. [Learn more](#)



I have 2 modifications:

- the 50kg pale malt are 25kg Golden Promise and 25kg Mar... ▾

Raw Farmhouse Harvest Ale Recipe (280L)

Batch Specifications

- **Final Volume:** 280 Liters (Watered down in Unitank)





- 25.0 kg Golden Promise Malt
- 25.0 kg Maris Otter Malt
- 5.0 kg Chit Malt
- 5.0 kg Cane Sugar (Added to the 2L Chinook bittering boil)
- Additional cane sugar used within the yeast starters

Hop Bill (Total: 250g Pellets + 2.24 kg Home-Dried Leaf)

- 250g Chinook Pellets (Boiled separately in 2L water)
- 450g Fuggie (Home-Dried Whole Leaf)
- 450g Willamette (Home-Dried Whole Leaf)
- 340g Styrian Golding (Home-Dried Whole Leaf)
- 340g Bramling Cross (Home-Dried Whole Leaf)
- 220g Phoenix (Home-Dried Whole Leaf)

Yeast & Nutrients

- Co-pitch: Hornindal Kveik + Ebbegarden Kveik
- Servomyces Yeast Nutrient (Added directly to the unitank)

Brew Day Process

1. Hop Preparation (Day Before)

- Harvest fresh hop cones in the morning after dew evaporates.
- Load 2.24 kg of fresh hops into the food dehydrator at 35°C–40°C.
- Dry for 12–18 hours until the central strigs are brittle and snap easily.

2. Mashing

- Perform your standard mash using the 25 kg Golden Promise, 25 kg Maris Otter, and 5 kg Chit Malt.
- Transfer the sweet wort into the 190L boil kettle.
- **Crucial Step:** Immediately clean and sanitize the empty mash tun and its false bottom





- Bring the main 190L kettle to exactly 80°C. Do not boil.
- Add all 2.24 kg of your home-dried free-floating whole-leaf hops into the kettle.
- Maintain the temperature at 80°C for 1 hour to pasteurize the wort and extract volatile oils. Run a weak whirlpool at the end.

5. The Hopback Transfer

- Route the plumbing: Kettle Outlet → HopRocket (Empty acting as a strainer) → Riptide Pump → Cleaned Mash Tun Inlet.
- Transfer the wort and loose hops into the cleaned mash tun.
- Let the mixture sit undisturbed on the false bottom for 10 minutes to allow the 2.24 kg of cones to form a natural filter bed.

6. Transfer, Inline Chilling, and Dilution

- Route the plumbing from the mash tun outlet: Mash Tun Outlet → Riptide Pump → Plate Chiller → Unitank.
- Begin pumping the clear wort through the plate chiller to drop the temperature immediately to your pitching target (33°C–35°C) on its way to the unitank.
- As the liquid level drops in the mash tun, slowly rinse the hop bed using your extra boiled dilution water to sweep remaining sugars through the plate chiller.
- Dilute the final wort volume in the unitank to roughly 280 Liters.

7. Fermentation

- Confirm the unitank wort temperature is stable at 33°C–35°C.
- Add the Servomyces nutrient directly into the unitank.
- Co-pitch the Hornindal and Ebbegarden kveik strains.
- Maintain a fermentation temperature of 27°C–33°C to drive the attenuation down to 1.007.

You can now press **Ctrl+P** (or **Cmd+P** on Mac) to print or save this updated recipe as a **PDF**. Do you need any tips on **backwashing your plate chiller** to prevent the fine hop particles and raw ale proteins from clogging it post-transfer?





Raw Farmhouse Harvest Ale Recipe (280L)

Batch Specifications

- **Final Volume:** 280 Liters (Watered down in Unitank)
- **Kettle Volume:** 190 Liters
- **Target Original Gravity (OG):** 1.059 (*Slightly higher due to the extra malt*)
- **Target Final Gravity (FG):** 1.009 - 1.011 (*Caramel malt adds unfermentable sugars*)
- **Estimated ABV:** 6.3% - 6.5%
- **Process Type:** Raw Ale (No-Boil, 80°C Hop Stand)

Ingredients

Grain Bill & Sugars

- 25.0 kg Golden Promise Malt
- 25.0 kg Maris Otter Malt
- 5.0 kg Caramel Malt (50 EBC)
- 5.0 kg Chit Malt
- 5.0 kg Cane Sugar (Added to the 2L Chinook bittering boil)
- Additional cane sugar used within the yeast starters

Hop Bill (Total: 250g Pellets + 2.24 kg Home-Dried Leaf)

- 250g Chinook Pellets (Boiled separately in 2L water)
- 450g Fuggle (Home-Dried Whole Leaf)
- 450g Willamette (Home-Dried Whole Leaf)
- 340g Styrian Golding (Home-Dried Whole Leaf)
- 340g Bramling Cross (Home-Dried Whole Leaf)
- 220g Phoenix (Home-Dried Whole Leaf)

Yeast & Nutrients





- Load 2.24 kg of fresh hops into the food dehydrator at 35 °C–40 °C.
- Dry for 12–18 hours until the central strigs are brittle and snap easily.

2. Mashing

- Perform your standard mash using the 25 kg Golden Promise, 25 kg Maris Otter, 5 kg Caramel Malt, and 5 kg Chit Malt.
- Transfer the sweet wort into the 190L boil kettle.
- **Crucial Step:** Immediately clean and sanitize the empty mash tun and its false bottom so it is ready to act as a hopback later.

3. The Bittering Tea & Sugar Boil

- On the side, boil 2 Liters of water.
- Add the 250g of Chinook pellets and the 5 kg of cane sugar.
- Boil vigorously to force alpha-acid isomerization and dissolve the sugars. Set aside.

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- Bring the main 190L kettle to exactly 80°C. Do not boil.
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6. Transfer, Inline Chilling, and Dilution

- Route the plumbing from the mash tun outlet: Mash Tun Outlet → Riptide Pump → Plate Chiller → Unitank.
- Begin pumping the clear wort through the plate chiller to drop the temperature immediately to your pitching target (33°C–35°C) on its way to the unitank.





... from the fermenter and refrigeration tank.

- Maintain a fermentation temperature of 27°C–33°C to drive the fermentation to completion.

The addition of the **50 EBC caramel malt** will add a stunning deep amber color and a rich, light-caramel complexity that will perfectly complement the dark fruit flavors of the Bramling Cross and Phoenix hops.

You can now hit **Ctrl+P / Cmd+P** to save the final sheet. Let me know if you need to troubleshoot any other **hardware configurations** before your harvest!

AI responses may include mistakes. [Learn more](#)



The target original is still 1055. I just forgot to mention the caramel malt.

Raw Farmhouse Harvest Ale Recipe (280L)

Batch Specifications

- **Final Volume:** 280 Liters (Watered down in Unitank)
- **Kettle Volume:** 190 Liters
- **Target Original Gravity (OG):** 1.055
- **Target Final Gravity (FG):** 1.009 - 1.011 (*Caramel malt adds unfermentable dextrins*)
- **Estimated ABV:** 5.8% - 6.1%
- **Process Type:** Raw Ale (No-Boil, 80°C Hop Stand)

Ingredients

Grain Bill & Sugars

... 250g of ...





- 250g Chinook Pellets (Boiled separately in 2L water)
- 450g Fuggle (Home-Dried Whole Leaf)
- 450g Willamette (Home-Dried Whole Leaf)
- 340g Styrian Golding (Home-Dried Whole Leaf)
- 340g Bramling Cross (Home-Dried Whole Leaf)
- 220g Phoenix (Home-Dried Whole Leaf)

Yeast & Nutrients

- Co-pitch: Hornindal Kveik + Ebbegarden Kveik
- Servomyces Yeast Nutrient (Added directly to the unitank)

Brew Day Process

1. Hop Preparation (Day Before)

- Harvest fresh hop cones in the morning after dew evaporates.
- Load 2.24 kg of fresh hops into the food dehydrator at 35°C–40°C.
- Dry for 12–18 hours until the central strigs are brittle and snap easily.

2. Mashing

- Perform your standard mash using the 25 kg Golden Promise, 25 kg Maris Otter, 5 kg Caramel Malt, and 5 kg Chit Malt.
- Transfer the sweet wort into the 190L boil kettle.
- **Crucial Step:** Immediately clean and sanitize the empty mash tun and its false bottom so it is ready to act as a hopback later.

3. The Bittering Tea & Sugar Boil

- On the side, boil 2 Liters of water.
- Add the 250g of Chinook pellets and the 5 kg of cane sugar.
- Boil vigorously to force alpha-acid isomerization and dissolve the sugars. Set aside.

4. The 80°C Hop Stand





- Transfer the wort and loose hops into the cleaned mash tun.
- Let the mixture sit undisturbed on the false bottom for 10 minutes to allow the 2.24 kg of cones to form a natural filter bed.

6. Transfer, Inline Chilling, and Dilution

- Route the plumbing from the mash tun outlet: Mash Tun Outlet → Riptide Pump → Plate Chiller → Unitank.
- Begin pumping the clear wort through the plate chiller to drop the temperature immediately to your pitching target (33°C–35°C) on its way to the unitank.
- As the liquid level drops in the mash tun, slowly rinse the hop bed using your extra boiled dilution water to sweep remaining sugars through the plate chiller.
- Dilute the final wort volume in the unitank to roughly 280 Liters (adjusting dynamically to land on your target 1.055 OG).

7. Fermentation

- Confirm the unitank wort temperature is stable at 33°C–35°C.
- Add the Servomyces nutrient directly into the unitank.
- Co-pitch the Hornindal and Ebbegarden kveik strains.
- Maintain a fermentation temperature of 27°C–33°C to drive the fermentation to completion.

The numbers are perfectly adjusted back to your **1.055 target OG**. You are all set to go.

Whenever you are ready, you can hit **Ctrl+P / Cmd+P** to lock this into a **PDF**. Do you need any assistance with **fermentation timelines** or **kegging/bottling strategies** for this raw ale style?

AI responses may include mistakes. [Learn more](#)





Raw Farmhouse Harvest Ale Recipe (280L)

Batch Specifications

- **Final Volume:** 280 Liters (Watered down in Unitank)
- **Kettle Volume:** 190 Liters
- **Target Original Gravity (OG):** 1.055
- **Target Final Gravity (FG):** 1.009 - 1.011 (Caramel malt adds unfermentable dextrins)
- **Estimated ABV:** 5.8% - 6.1%
- **Process Type:** Raw Ale (No-Boil, 80°C Hop Stand)

Ingredients

Grain Bill & Sugars

- 25.0 kg Golden Promise Malt
- 25.0 kg Maris Otter Malt
- 5.0 kg Caramel Malt (50 EBC)
- 5.0 kg Chit Malt
- 5.0 kg Cane Sugar (Added to the 2L Chinook bittering boil)
- Additional cane sugar used within the yeast starters

Hop Bill (Total: 250g Pellets + 2.24 kg Home-Dried Leaf)

- 250g Chinook Pellets (Boiled separately in 2L water)
- 450g Fuggle (Home-Dried Whole Leaf)
- 450g Willamette (Home-Dried Whole Leaf)
- 340g Styrian Golding (Home-Dried Whole Leaf)
- 340g Bramling Cross (Home-Dried Whole Leaf)
- 220g Phoenix (Home-Dried Whole Leaf)

Yeast & Nutrients

- Co-nitch: Hornindal Kveik + Ebbeharden Kveik





- Load 2.24 kg of fresh hops into the food dehydrator at 35 °C–40 °C.
- Dry for 12–18 hours until the central strigs are brittle and snap easily.

2. Mashing

- Perform your standard mouthfeel mash using the 25 kg Golden Promise, 25 kg Maris Otter, 5 kg Caramel Malt, and 5 kg Chit Malt.
- Transfer the sweet wort into the 190L boil kettle.
- **Crucial Step:** Immediately clean and sanitize the empty mash tun and its false bottom so it is ready to act as a hopback later.

3. The Bittering Tea & Sugar Boil

- On the side, boil 2 Liters of water.
- Add the 250g of Chinook pellets and the 5 kg of cane sugar.
- Boil vigorously to force alpha-acid isomerization and dissolve the sugars. Set aside.

4. The 80°C Hop Stand

- Bring the main 190L kettle to exactly 80°C. Do not boil.
- Add all 2.24 kg of your home-dried free-floating whole-leaf hops into the kettle.
- Maintain the temperature at 80°C for 1 hour to pasteurize the wort and extract volatile oils. Run a weak whirlpool at the end.

5. The Hopback Transfer

- Route the plumbing: Kettle Outlet → HopRocket (Empty acting as a strainer) → Riptide Pump → Cleaned Mash Tun Inlet.
- Transfer the wort and loose hops into the cleaned mash tun.
- Let the mixture sit undisturbed on the false bottom for 10 minutes to allow the 2.24 kg of cones to form a natural filter bed.

6. Transfer, Inline Chilling, and Dilution

- Route the plumbing from the mash tun outlet: Mash Tun Outlet → Riptide Pump → Plate Chiller → Unitank.
- Begin pumping the clear wort through the plate chiller to drop the temperature immediately to your pitching target (33°C–35°C) on its way to the unitank.





... from the fermenter and refrigeration system.

- Maintain a fermentation temperature of 27°C–33°C to drive the fermentation to completion.

Recipe Co-Created By:

- **Head Brewer:** You 🌾
- **Brewing Assistant:** Google AI 🤖

Skål and happy brewing! May your fermentation be vigorous and your harvest exceptionally aromatic.

Best of luck with the harvest and brew day! When the beer is finished and carbonated, reach back out if you want to brainstorm a creative **name and label concept** based on its unique **Anglo-Scandinavian farmhouse heritage**.

AI responses may include mistakes. [Learn more](#)



Sweet! Ohh, I forgot... the brewery is Keldling Brew, it is brew #187 and the beer title is 'Pick Me!'... maybe that should be ... ▾

Raw Farmhouse Harvest Ale Recipe (280L)

Brewery: Keldling Brew
Brew: #187
Beer Title: 'Pick Me!'

Batch Specifications

- **Final Volume:** 280 Liters (Watered down in Unitank)
- **Kettle Volume:** 190 Liters





- 25.0 kg Golden Promise Malt
- 25.0 kg Maris Otter Malt
- 5.0 kg Caramel Malt (50 EBC)
- 5.0 kg Chit Malt
- 5.0 kg Cane Sugar (Added to the 2L Chinook bittering boil)
- Additional cane sugar used within the yeast starters

Hop Bill (Total: 250g Pellets + 2.24 kg Home-Dried Leaf)

- 250g Chinook Pellets (Boiled separately in 2L water)
- 450g Fuggle (Home-Dried Whole Leaf)
- 450g Willamette (Home-Dried Whole Leaf)
- 340g Styrian Golding (Home-Dried Whole Leaf)
- 340g Bramling Cross (Home-Dried Whole Leaf)
- 220g Phoenix (Home-Dried Whole Leaf)

Yeast & Nutrients

- Co-pitch: Hornindal Kveik + Ebbegarden Kveik
- Servomyces Yeast Nutrient (Added directly to the unitank)

Brew Day Process

1. Hop Preparation (Day Before)

- Harvest fresh hop cones in the morning after dew evaporates.
- Load 2.24 kg of fresh hops into the food dehydrator at 35°C–40°C.
- Dry for 12–18 hours until the central strigs are brittle and snap easily.

2. Mashing

- Perform your standard mouthfeel mash using the 25 kg Golden Promise, 25 kg Maris Otter, 5 kg Caramel Malt, and 5 kg Chit Malt.
- Transfer the sweet wort into the 100L boil kettle





4. The 80°C Hop Stand

- Bring the main 190L kettle to exactly 80°C. Do not boil.
- Add all 2.24 kg of your home-dried free-floating whole-leaf hops into the kettle.
- Maintain the temperature at 80°C for 1 hour to pasteurize the wort and extract volatile oils. Run a weak whirlpool at the end.

5. The Hopback Transfer

- Route the plumbing: Kettle Outlet → HopRocket (Empty acting as a strainer) → Riptide Pump → Cleaned Mash Tun Inlet.
- Transfer the wort and loose hops into the cleaned mash tun.
- Let the mixture sit undisturbed on the false bottom for 10 minutes to allow the 2.24 kg of cones to form a natural filter bed.

6. Transfer, Inline Chilling, and Dilution

- Route the plumbing from the mash tun outlet: Mash Tun Outlet → Riptide Pump → Plate Chiller → Unitank.
- Begin pumping the clear wort through the plate chiller to drop the temperature immediately to your pitching target (33°C–35°C) on its way to the unitank.
- As the liquid level drops in the mash tun, slowly rinse the hop bed using your extra boiled dilution water to sweep remaining sugars through the plate chiller.
- Dilute the final wort volume in the unitank to roughly 280 Liters (adjusting dynamically to land on your target 1.055 OG).

7. Fermentation

- Confirm the unitank wort temperature is stable at 33°C–35°C.
- Add the Servomyces nutrient directly into the unitank.
- Co-pitch the Hornindal and Ebbegarden kveik strains.
- Maintain a fermentation temperature of 27°C–33°C to drive the fermentation to completion.

Recipe Co-Created By:

... .. 🍷





AI responses may include mistakes. [Learn more](#)



Ohh no, I forgot the details with pH control. I add Lime/Lemon juice and/or Citric acid to lower the pH to 5.0 before transfer... ▾

Raw Farmhouse Harvest Ale Recipe (280L)

Brewery: Keldling Brew

Brew: #187

Beer Title: 'Pick Me!'

Batch Specifications

- **Final Volume:** 280 Liters (Watered down in Unitank)
- **Kettle Volume:** 190 Liters
- **Target Original Gravity (OG):** 1.055
- **Target Final Gravity (FG):** 1.009 - 1.011 (Caramel malt adds unfermentable dextrins)
- **Estimated ABV:** 5.8% - 6.1%
- **Process Type:** Raw Ale (No-Boil, 80°C Hop Stand)

Ingredients

Grain Bill & Sugars

- 25.0 kg Golden Promise Malt
- 25.0 kg Maris Otter Malt
- 5.0 kg Caramel Malt (50 EBC)
- 5.0 kg Chit Malt
- 5.0 kg Cane Sugar (Added to the 2L Chinook bittering boil)
- Additional cane sugar used within the yeast starters





Water Adjustments & pH Control

- Lime juice, Lemon juice, and/or Citric Acid (To hit target transfer pH)

Yeast & Nutrients

- Co-pitch: Hornindal Kveik + Ebbegarden Kveik
- Servomyces Yeast Nutrient (Added directly to the unitank)

Brew Day Process

1. Hop Preparation (Day Before)

- Harvest fresh hop cones in the morning after dew evaporates.
- Load 2.24 kg of fresh hops into the food dehydrator at 35°C–40°C.
- Dry for 12–18 hours until the central strigs are brittle and snap easily.

2. Mashing

- Perform your standard mouthfeel mash using the 25 kg Golden Promise, 25 kg Maris Otter, 5 kg Caramel Malt, and 5 kg Chit Malt.
- Transfer the sweet wort into the 190L boil kettle.
- **Crucial Step:** Immediately clean and sanitize the empty mash tun and its false bottom so it is ready to act as a hopback later.

3. The Bittering Tea & Sugar Boil

- On the side, boil 2 Liters of water.
- Add the 250g of Chinook pellets and the 5 kg of cane sugar.
- Boil vigorously to force alpha-acid isomerization and dissolve the sugars. Set aside.

4. The 80°C Hop Stand & pH Adjustment

- Bring the main 190L kettle to exactly 80°C. Do not boil.
- Add all 2.24 kg of your home-dried free-floating whole-leaf hops into the kettle.
- Maintain the temperature at 80°C for 1 hour to pasteurize the wort and extract volatile





- Transfer the 5.0 pH wort and loose hops into the cleaned mash tun.
- Let the mixture sit undisturbed on the false bottom for 10 minutes to allow the 2.24 kg of cones to form a natural filter bed.

6. Transfer, Inline Chilling, and Dilution

- Route the plumbing from the mash tun outlet: Mash Tun Outlet → Riptide Pump → Plate Chiller → Unitank.
- Begin pumping the clear wort through the plate chiller to drop the temperature immediately to your pitching target (33°C–35°C) on its way to the unitank.
- As the liquid level drops in the mash tun, slowly rinse the hop bed using your extra boiled dilution water to sweep remaining sugars through the plate chiller.
- Dilute the final wort volume in the unitank to roughly 280 Liters (adjusting dynamically to land on your target 1.055 OG).

7. Fermentation

- Confirm the unitank wort temperature is stable at 33°C–35°C.
- Add the Servomyces nutrient directly into the unitank.
- Co-pitch the Hornindal and Ebbegarden kveik strains.
- Maintain a fermentation temperature of 27°C–33°C to drive the fermentation to completion.

Recipe Co-Created By:

- **Head Brewer:** Keldling Brew 🍷
- **Brewing Assistant:** Google AI 🤖

Skål to Brew #187! May 'Pick Me!' be your best harvest ale yet.

The pH step is now fully locked into **Section 4**. Go ahead and hit **Ctrl+P** / **Cmd+P** to export the final PDF sheet. Let me know if there are any other **water chemistry** or **process notes** you want to drop in before you heat up the strike water!



